Building a Smart Financial Data Assistant

In today's financial and telecom industries, organizations handle a large number of customer credit applications every day. Making informed lending decisions requires a clear understanding of customer behavior, borrowing patterns, and potential risk factors. The goal of this project was to analyze customer credit data and build a Smart Financial Data Assistant that helps business users understand the data through dashboards and natural language questions.

For this project, I used the German Credit dataset, which contains information about 1,000 customers. The dataset includes customer details such as age, housing status, loan amount, loan duration, savings accounts, and loan purpose. The objective was to identify meaningful patterns within the data and present them in a way that supports business decision-making.

The first step involved assessing the quality of the dataset. Missing values were identified in a few financial attributes, particularly savings accounts and checking accounts. The data was then cleaned and transformed to make it easier to analyze. Additional categories such as age groups and credit bands were created to improve the analysis and visualization process.

After preparing the data, exploratory analysis was performed to understand customer behavior and identify key trends. One of the most noticeable findings was that customers living in free housing tend to request higher credit amounts compared to customers who own or rent their homes. These customers also showed longer average loan durations, which may indicate a higher level of financial exposure.

The analysis also revealed that car loans and radio/TV loans make up the largest share of loan applications. This suggests that a significant portion of the customer portfolio is concentrated within a small number of loan categories. Understanding these patterns can help organizations monitor their lending portfolio more effectively.

Another important observation was the age distribution of customers. Most applicants belong to the 26–50 age group, making it the largest customer segment in the dataset. This information can be useful when designing products, services, and risk strategies targeted toward the organization's primary customer base.

A strong positive relationship was also observed between credit amount and loan duration. Customers requesting larger loans generally require longer repayment periods. From a business perspective, these loans may require additional monitoring because they increase financial exposure over time.

To communicate these findings effectively, an interactive Power BI dashboard was developed. The dashboard includes an Executive Overview page for high-level business insights and a Risk Analysis page that focuses on factors influencing customer exposure. Interactive filters allow users to explore the data from different perspectives and gain a deeper understanding of customer behavior.

As part of the project, a Smart Financial Data Assistant prototype was also created. The assistant allows users to ask business-related questions in simple English and receive structured responses based on the dataset. It can answer questions related to loan purposes, customer segments, housing categories, and loan characteristics. The assistant also explains the data used to generate responses and clearly identifies questions that cannot be answered due to limitations in the available dataset.

Based on the analysis, organizations should pay closer attention to customers requesting large loan amounts with longer repayment durations. Additional customer information such as income, repayment history, and previous credit performance would further improve future risk assessments and decision-making processes.

Overall, this project demonstrates how data analysis, visualization, and conversational AI can work together to transform raw financial data into actionable business insights. The solution provides a foundation for building more advanced analytics and decision-support systems in real-world financial and telecom environments.

AtLast with the attachments and all we come to the execution of end task

GITHUB REPO :- https://github.com/khadeshri19/Amdocs_Assignment
VERCEL DEPLOPYOD LINK :- https://chat-bot-amdocsassignment.vercel.app/